

WATSONVILLE, CA, USA

WMO: 745058

Lat: 36.936N

Lon: 121.789W

Elev: 49

StdP: 100.74

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 23277

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.2	2.4	-5.6	2.4	11.4	-3.4	2.9	10.1	8.2	13.5	7.0	13.8	1.1	330	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	11.4	30.1	16.9	27.3	16.7	25.0	16.2	18.6	26.5	17.8	25.1	17.1	23.6	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.5	19.2	15.2	10.8	19.0	14.6	10.4	18.5	52.4	26.7	50.0	25.0	48.2	23.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	5.5	4.9	-1.0	36.0	1.4	2.4	-2.1	37.7	-2.9	39.1	-3.7	40.4	-4.8	42.2
			-2.1	20.5	1.4	0.8	-3.1	21.1	-3.9	21.6	-4.7	22.0	-5.7	22.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	10.7	11.3	12.3	13.1	14.7	16.2	17.0	17.4	17.7	16.2	13.1	10.5
	DBStd	3.57	2.72	2.76	2.61	2.59	2.17	1.99	1.72	1.93	2.74	2.88	2.83	2.56
	HDD10.0	82	25	16	7	3	0	0	0	0	0	0	6	25
	HDD18.3	1586	236	198	187	159	118	69	51	42	45	79	159	243
	CDD10.0	1614	48	52	79	96	144	187	216	230	229	193	98	41
	CDD18.3	76	0	0	1	3	3	6	8	13	25	15	2	0
	CDH23.3	918	6	8	22	51	62	88	68	101	256	217	38	1
	CDH26.7	305	1	0	3	18	19	28	11	29	107	81	7	0
Wind	WSAvg	1.8	1.5	1.8	1.9	2.1	2.2	2.2	2.1	2.0	1.8	1.6	1.4	1.5
Precipitation	PrecAvg	542	107	89	91	48	5	3	2	2	7	29	84	90
	PrecMax	1228	278	265	280	169	25	33	32	28	37	97	204	266
	PrecMin	271	6	5	0	0	0	0	0	0	0	0	2	1
	PrecStd	207	74	75	62	50	6	6	7	5	11	28	61	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	24.5	26.3	28.9	29.0	30.4	27.9	30.3	34.1	32.8	27.5	22.0
		MCWB	12.5	14.9	16.7	17.2	17.2	17.7	18.5	18.5	18.2	16.7	15.7	12.0
	2%	DB	20.8	21.7	22.5	23.7	24.3	25.7	25.2	25.9	29.8	28.6	23.6	18.9
		MCWB	12.6	13.6	14.7	14.9	15.4	16.9	17.5	17.7	16.6	15.7	14.6	12.0
	5%	DB	17.9	18.5	19.7	20.6	21.5	22.9	23.0	23.4	26.0	25.2	20.9	17.1
		MCWB	12.1	12.5	13.4	13.5	14.9	16.0	16.7	17.0	16.9	15.0	13.3	11.8
	10%	DB	15.8	16.3	17.5	18.1	19.4	21.2	21.3	21.9	22.8	22.2	18.1	15.3
		MCWB	11.5	11.9	12.6	13.0	14.1	15.4	16.1	16.5	16.3	14.5	12.7	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	16.2	17.4	17.9	18.3	18.8	19.1	19.1	19.6	18.6	17.4	16.0
		MCDB	18.9	21.3	25.0	27.7	26.5	27.6	26.7	27.5	28.0	27.0	23.4	17.9
	2%	WB	14.0	14.9	15.6	15.9	16.3	17.4	17.8	18.0	18.3	17.3	15.9	14.6
		MCDB	17.1	19.1	20.7	22.1	22.9	24.6	24.5	24.8	26.3	24.5	20.0	16.1
	5%	WB	13.1	13.7	14.4	14.6	15.2	16.4	16.9	17.3	17.4	16.4	14.9	13.4
		MCDB	16.0	17.3	18.1	19.1	20.4	22.4	22.6	23.2	24.5	22.5	18.7	15.1
	10%	WB	12.2	12.6	13.3	13.5	14.4	15.6	16.2	16.6	16.6	15.4	13.9	12.3
		MCDB	14.9	15.5	16.5	17.3	18.9	20.6	21.0	21.5	22.3	20.6	17.3	14.3
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.1	10.4	10.6	10.3	10.4	9.3	9.4	11.4	12.3	11.6	10.3
		MCDBR	14.5	14.8	14.9	15.5	14.9	14.7	12.8	12.9	17.3	18.2	15.9	12.9
		MCWBR	8.2	8.2	8.3	8.3	7.7	7.1	6.0	6.0	7.3	7.8	7.9	7.4
	5% WB	MCDBR	10.8	11.4	12.2	12.9	12.8	12.8	11.8	12.1	14.4	14.2	11.8	7.8
		MCWBR	6.8	6.9	7.4	7.5	7.0	6.4	5.6	5.6	6.4	6.7	6.6	5.4
Clear-Sky Solar Irradiance	taub		0.319	0.325	0.326	0.353	0.364	0.355	0.358	0.342	0.350	0.323	0.311	0.315
	taud		2.548	2.538	2.530	2.431	2.416	2.452	2.461	2.496	2.465	2.576	2.576	2.574
	Ebn at Noon		876	915	943	928	917	922	916	928	903	904	879	857
	Edh at Noon		77	88	97	113	116	112	110	104	102	83	74	71
All-Sky Solar Radiation	RadAvg		2.55	3.45	4.87	6.21	7.18	7.84	7.39	6.66	5.68	4.23	2.97	2.25
	RadStd		0.30	0.39	0.41	0.52	0.42	0.32	0.22	0.25	0.21	0.25	0.16	0.29

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.36	N/A	N/A	N/A	N/A	+0.70	N/A	-195	+184	N/A
(47)	Regional (44 neighbors)	+0.37	N/A	-1.22	N/A	N/A	N/A	N/A	-108	+136	+31

Nomenclature: See separate page